Review of Machine Learning Approaches for Diagnostics and Prognostics of Industrial Systems Using Industrial Open Source Data

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Abstract

In the field of Prognostics and Health Management (PHM), recent years have witnessed a significant surge in the application of machine learning (ML). Despite this growth, the field grapples with a lack of unified guidelines and systematic approaches for effectively implementing these ML techniques and comprehensive analysis regarding industrial open-source data across varied scenarios. To address these gaps, this paper provides a comprehensive review of machine learning approaches for diagnostics and prognostics of industrial systems using open-source datasets from PHM Data Challenge Competitions held between 2018 and 2023 by PHM Society and IEEE Reliability Society and summarizes a unified ML framework. This review systematically categorizes and scrutinizes the problems, challenges, methodologies, and advancements demonstrated in these competitions, highlighting the evolving role of both conventional machine learning and deep learning in tackling complex industrial tasks related to detection, diagnosis, assessment, and prognosis. Moreover, this paper delves into the common challenges in PHM data challenge competitions by emphasizing both data-related and model-related issues and summarizes the solutions that have been employed to address these challenges. Finally, we identify key themes and potential directions for future research, providing opportunities and prospects for ML further development in PHM.

1 Introduction

In the era of Industry 4.0, the emphasis on the reliability, efficiency, and longevity of industrial systems has become crucial [53]. Against this backdrop, Prognostics and Health Management (PHM) has emerged as an indispensable discipline, that integrates the detection, diagnosis, assessment, and prognosis of system failures addressing the growing need for proactive system health management [127]. This integrative approach can bolster the reliability and safety of industrial systems, with minimal unplanned downtimes and reduced maintenance costs.

The rise of the Internet of Things [94], big data analytics [110], cyber-physical systems [55], machine learning (ML) [37], deep learning (DL) [87], and industrial artificial intelligence [81, 57] has paved the way for a transformative shift in PHM. Historically, PHM strategies largely relied on physical-based methods, which derived their strength from profound insights into system physics, material properties, and failure mechanisms. These approaches, however, often struggled with scalability, adaptability, and the ability to handle the vast variability and uncertainties inherent in real-world operations. In contrast, ML techniques present the capability to model complex systems comprehensively, uncover intricate patterns, and accurately diagnose and predict failures. Yet, employing ML in PHM poses several challenges, including the critical need for usable, useful, high-quality, and extensive data sets, extensive computing resources, a solid infrastructure for data collection and processing, as well as the necessity for assembly of a team of professionals proficient in artificial intelligence (AI) and ML [56, 5]. Despite these challenges, the momentum for implementing ML in industrial settings is evident. With the gradual move towards digital transformation within the industry, incorporating ML into PHM aligns with the principles of Industry 4.0 [82]. It underscores the capability of data-driven approaches to provide precise and adaptable diagnostic and prognostic solutions and represents a strategic shift towards leveraging ML techniques for enhanced predictive maintenance.

1.1 A Survey of Relevant Reviews

A multitude of comprehensive reviews on ML strategies for PHM are noted [90, 83, 52, 79, 122, 78, 123, 82, 87]. These reviews evaluate the existing literature both qual-

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itatively and quantitatively, presenting their distinct perspectives and pinpointing the trends and new concepts of ML strategies for PHM across various scenarios. [90] reviewed DL architectures like a one-class neural network (OCNN), self-organizing map (SOM), and generative techniques aligned with industrial needs from a predictive maintenance perspective. [83] highlighted seven DL architectures, including emerging DL methods such as graph neural networks, transformers, and generative adversarial networks, addressing four different challenges (imbalanced data, multimodal data fusion, compound fault types, and edge device implementation). Additionally, [52] provided a comprehensive examination of PHM strategies in the context of smart factories, spanning from traditional approaches to DL-based approaches. An extensive review on modeling techniques supporting PHM of industrial equipment, specifically within onshore wind energy and civil aviation sectors, was given by [122], wherein they discuss how modeling approaches are shaped by industry-specific factors (maintenance strategies, implementation aspects, and supporting technologies). Furthermore, [78] proposed a general guideline under AI-based PHM for selecting appropriate techniques to solve specific PHM problems.

The aforementioned review papers serve as the groundwork for further PHM development. Currently, a significant portion of the literature primarily examines algorithms based on their capabilities and functionalities, with numerous reviews covering various ML or DL methods like CNN, RNN, GAN, GNN, Transformer, etc. However, how these algorithms are applied within actual industrial settings is still rarely considered. The practice among many researchers of relying on artificial data sets for algorithm testing, instead of using real, industry-specific data sets, is notable. Furthermore, a considerable portion of the data sets mentioned in these studies are not publicly accessible, and among the available ones, most date back more than six years (before 2018). Additionally, the focus of discussed data sets predominantly lies on mechanical components commonly used in algorithm development, such as gears and bearings. In light of these observations, this paper will try to analyze the open source data sets available in the last six years and conduct an in-depth analysis from the perspective of problem-challenge-solution-application orientation.

1.2 Motivation

Over the last fifteen years, organizations such as the PHM Society and the IEEE Reliability Society have played pivotal roles in fostering innovation, research, and collaboration in the PHM field. One notable initiative is the PHM data challenge competition, which stands as a testament to the growing emphasis on real-world, data-centric PHM solutions. By providing participants with real industrial datasets and posing real-world challenges, this competition seeks to accelerate the development and validation of cutting-edge PHM methodologies, bridging the gap between academia and industry. In light of the recent surge in data-driven methodologies, particularly ML, and DL, in PHM, it becomes essential to revisit and scrutinize research algorithms and methodologies derived from the PHM data challenge competitions and their datasets. Systematically reviewing the challenges presented by industrial data across various sectors with methodologies and analytics can foster and enhance a deeper understanding of data-driven approaches. Such endeavors not only propel the evolution of data-centric techniques, ML, and AI within the PHM domain but also accentuate advancements in the broader landscape of industrial data analytics and industrial artificial intelligence.

After evaluating competitions over the past 15 years, we chose to concentrate on the competitions held after 2017, specifically from 2018 to 2023, covering a span of six years. This decision is motivated by several factors. Firstly, as we mentioned in section 1.1, the open-source datasets that are utilized for analytics in the aforementioned review papers are most before 2018, while the post-2018 open-source datasets are rarely discussed in those reviews. Secondly, the PHM data challenge competition offers open-source datasets, facilitating deeper investigation by researchers into the challenges prevalent across various industrial settings. This, in turn, encourages the development of innovative algorithms and fosters advancements in the PHM field. Thirdly, recent years have witnessed rapid advancements and iterations in ML and DL technologies, leading to the emergence of a large number of complex architectures and algorithms. Reviewing these papers can help to better understand the latest trends of ML and DL in PHM and their role in the future development of PHM.

1.3 Contributions

This study aims to conduct a comprehensive problem-challenge-solution-application-oriented review within the domain of PHM Data Challenge. In pursuit of the objective, approximately sixty research papers were reviewed, encompassing both competition winning contributions and subsequent exploratory research undertaken post-competition. Our contributions include the following:

1. This study summarizes the problems and solutions presented in nine PHM data challenge competitions, elucidating the ML or DL algorithms, methods, and analytical strategies employed to tackle these competitions.

2. We propose a unified ML framework for the PHM do-

Table 1. Overview of PHM Data Challenge Competitions from 2018 to 2023

Competition	Industrial Systems	Research Task
2018 PHM NA	Ion Mill Etching System	Detection & Diagnosis & Assessment & Prognosis
2019 PHM NA	Fatigue Crack	Assessment & Prognosis
2020 PHM EU	Filtration System	Prognosis
2021 PHM EU	Manufacturing Production Line	Detection & Diagnosis
2021 PHM NA	Turbofan Engine	Prognosis
2022 PHM EU	Printed Circuit Board	Detection & Diagnosis
2022 PHM NA	Rock Drill	Detection & Diagnosis
2023 PHM AP	Spacecraft Propulsion System	Detection & Diagnosis
2023 IEEE	Gearbox	Detection & Diagnosis
Common Challenges: Data Imbalance, Domain Shift, Multiple Fault Modes, Unknown Fault, Limited Data Availability		

main based on this review study, serving as a general guideline for the development of future ML models.

3. We discuss common challenges associated with industrial data, underscoring specific issues related to data issues (missing data, data imbalance, and domain shift), and model issues (model selection, machine learning model interpretability, model robustness and generalization) in data-driven approaches. Possible solutions are also provided. Moreover, we evaluate the limitations of these competitions and suggest future directions for their evolution.

4. We identify six further research directions for the application of ML in PHM. These include (1) a need for open-source multi-modal datasets, (2) development of multi-modal machine learning approaches, (3) further exploration in machine learning model interpretability, (4) novel transfer learning and domain adaptation techniques development for model robustness and generalization, (5) construction of large knowledge library, and (6) potential utilization of large language models and industrial large knowledge models.

The rest of this paper is organized as follows: Section 2 provides an overview of PHM data challenge competitions, a unified ML framework for the PHM domain, and underscores the principal research tasks within PHM. Section 3 and Section 4 respectively introduce the prevalent challenges associated with two pivotal tasks: detection & diagnosis, and assessment & prognosis, subsequently detailing the solutions presented in various PHM data competitions individually. Section 5 critically summarizes and examines common challenges in the PHM domain through the perspective of ML methodologies, addressing concerns related to data issues, and model issues. The limitations of these competitions are also examined. Section 6 provides a forward-looking perspective on the enhanced integration of data-driven techniques, ML, DL, and AI in future PHM development. Section 7 concludes the paper, highlighting its findings and contributions.

2 Background

In this section, we begin with an overview of the PHM data challenge competition. Next, we outline the core research tasks in the PHM field and provide their respective explanations. We then summarize the issues and predominant challenges posed by various competitions and review the data-driven approaches used in the competitions.

2.1 Overview of PHM Data Challenge Competitions

From 2018 to 2023, PHM Society and IEEE Reliability Society have initiated nine PHM data challenge competitions which reflect challenges associated with analyzing industrial data across diverse industrial sectors. The main purpose behind these competitions is to amplify focus and collaborative efforts from both academic and industrial realms to address real-world challenges. The topics of these competitions are commendably extensive, encompassing diverse industrial systems such as Ion Mill

Etching Tools, Filtration Systems, Manufacturing Production Lines, Turbofan Engines, Printed Circuit Boards, Rock Drills, Spacecraft Propulsion Systems, and Gearbox, among others. Furthermore, the range of problems posed encapsulates the main tasks in PHM domain including detection, diagnosis, assessment, and prognosis. These tasks align with PHM's ultimate goal: to accurately forecast system health, degradation, and eventual failure, thereby improving system reliability, safety, and operational efficiency.

For ease of discussing different competitions, we've adopted a condensed naming convention for the PHM data challenge competition: "YEAR ORGANIZATION" indicates the respective year and organizing body of the challenge. Among the challenges discussed, one is organized by the IEEE Reliability Society, while the remaining eight are organized by the PHM Society. The PHM Society conducts an annual conference in North America, an Asia-Pacific conference in odd years, and a European conference in even years. We use the abbreviations "PHM NA", "PHM AP", and "PHM EU" to represent the competitions held in North America, Asia-Pacific, and Europe, respectively. For instance, "2018 PHM NA" refers to the challenge held in North America by the PHM Society in 2018 while "2023 IEEE" represents the competition hosted by the IEEE Reliability Society in 2023. Table 1 presents the systems, research tasks, and common challenges of the nine data competitions. For an in-depth overview of the PHM data challenge competitions and their associated datasets, please refer to Appendix, Section 3 and Section 4.

2.2 Core Research Tasks within PHM

In the realm of PHM, there are four pivotal research tasks: detection, diagnosis, assessment, and prognosis.

Detection: The foundational step of PHM. It entails continuous monitoring of equipment and systems to identify any anomalies or deviations from expected performance. Notably, the primary aim of this phase is anomaly identification (detection) rather than root cause analysis.

Diagnosis: Upon detecting anomalies, the diagnostic phase delves deeper to determine failure types or failure modes and may find out the root causes of the problem, enabling targeted corrective actions in a timely manner. In diagnostic scenarios, detected failures often need to be classified into specific failure types.

Assessment: In the assessment phase, the current operational status and performance of the system will be evaluated. Leveraging either historical data or recent machinery behavior, this phase evaluates potential risks or assesses the health status of the system in its present condition.

Prognosis: Representing the most forward-looking phase, prognosis leverages both current and historical data to forecast the future health of a system or the residual life of a machine. Commonly, this is referred to as predicting the Remaining Useful Life (RUL). This pivotal phase provides estimates on potential system or machinery failure timelines, thereby facilitating proactive maintenance planning. To achieve this, data-driven approaches, including ML and AI, are often employed to construct predictive models for future prognosis.

Reflecting on the nine data competitions spanning the last six years, Figure 1 illustrates the associated research tasks. Our analysis yielded the following insights:

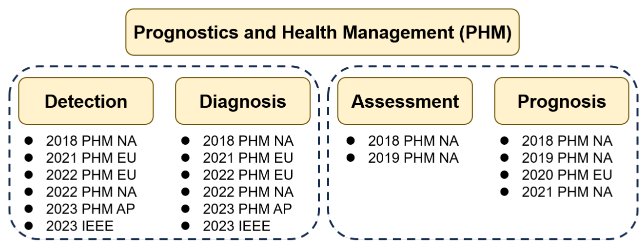

Figure 1. Research Tasks in PHM Data Challenge Competitions

Integration of Fault Detection and Diagnosis: Fault detection and diagnosis frequently co-occur in research tasks. While fault detection primarily identifies discrepancies or anomalies compared with expected performance, it doesn't delve into their root causes or subsequent maintenance implications. Following fault detection, diagnosis intensively examines the root causes of the faults, categorizing them by distinct fault types. This categorization aids in guiding subsequent maintenance operations.

Synergy between Health Assessment and Prognosis: There exists an intrinsic link between tasks centered on health assessment and those focused on prognosis. While health assessment provides the current state of system health, prognostic tasks often predict future states based on the current assessment. Notably, even in competitions or tasks primarily targeting prognosis, emphasis is occasionally placed initially on health assessment methodologies. This precedence stems from the understanding that a robust health assessment forms the foundation for efficacious prognostic strategies. Recognizing a system's current health state equips researchers with a more potent toolkit to tackle prognostic challenges.

2.3 Comprehensive Review of Data-Driven Approaches in Recent PHM Data Challenge Competitions

After conducting an in-depth review of data competitions from the last six years, Figure 3 illustrates the publication count and the density of specific ML or DL approaches dis-

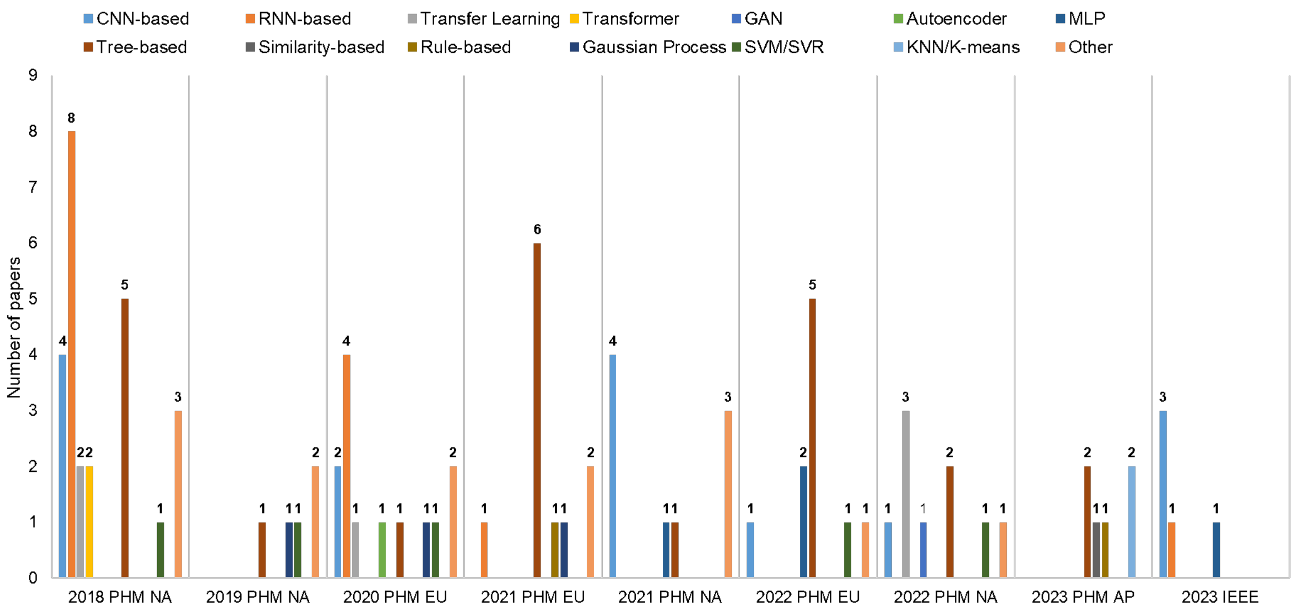

Figure 2. Distribution of ML Approaches Across PHM Data Challenge Competitions

cussed in this paper. Furthermore, Figure 2 provides details into the frequency of particular ML approaches within each competition. It is important to note that we count the occurrences of distinct ML methods mentioned in a research paper. Given that some articles employ multiple ML methods, the aggregate count of methods exceeds the total number of publications.

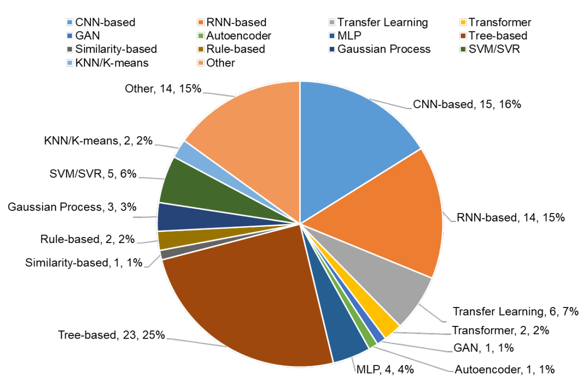

Figure 3. The Density of Various ML Approaches in PHM Data Challenge Competitions from 2018 until December 2023

Moreover, we have summarized a unified ML framework that concludes ML approaches for PHM data competitions during this period. As shown in Figure 4, it encompasses five primary components: Data Collection, Data Processing, Data Visualization, Conventional Machine Learning & Deep Learning, and Model Interpretability.

Data Collection: Data collection is the foundation of these open-source data challenge competitions (PHM Society[1] and IEEE Reliability Society[2]). Companies across various industries contribute datasets with the objective of harnessing data-driven strategies to address intricate challenges in PHM. Typically, these industrial datasets encompass a diverse range of data types, including but not limited to time-series data, tabular data, images, sensor readings, and simulation data.

Data Processing: A pivotal stage before ML model development is data processing, which is crucial for enhancing data quality and ensuring effective model training [17, 105]. This stage can be further subdivided into two aspects: data preprocessing and feature engineering. Data preprocessing addresses raw data challenges, typically including missing data, noise, outliers, data imbalance, and scaling issues. Techniques like imputation [28], denoising [22], outlier detection [72], resampling [16], and normalization & standardization [54, 33] are commonly applied. Feature engineering follows, refining data representation post-preprocessing. This step often uncovers hidden insights and deepens understanding of data, therefore significantly enhancing model performance and predictive capabilities in the later stage [92].

Data Visualization: Data visualization is another important aspect of the ML process. It involves transforming data into intuitive graphical representations, such as graphs

[1]`https://phmsociety.org/`
[2]`https://rs.ieee.org/`

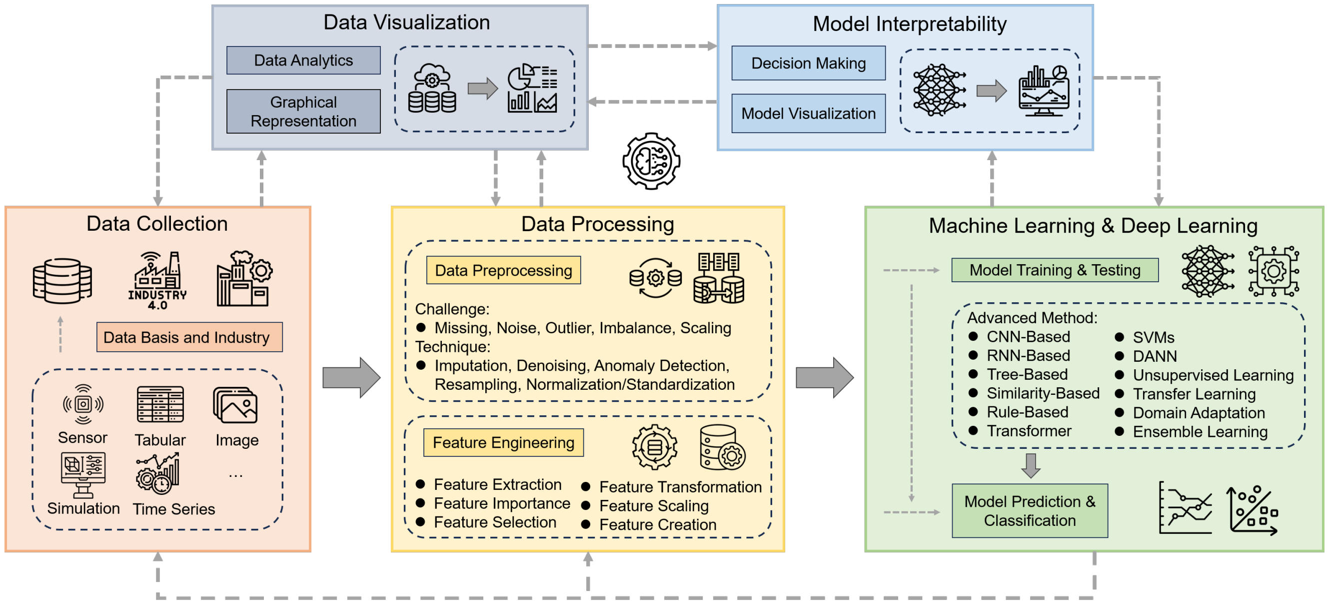

Figure 4. A Unified Machine Learning Framework in Prognostics and Health Management Domain

or charts. Researchers can use these charts to better observe trends, patterns, or outliers in data, which further help people have a better understanding of data and generate some useful insights. In the PHM domain, effective visualization can help initial data exploration and analysis, and accelerate data processing and ML modeling development [12, 15].

Conventional Machine Learning & Deep Learning Techniques: Post data processing, various conventional ML, and DL methods are developed to solve the competition problems using useful and usable data, as shown in Figure 4. This segment includes "model training and testing" and "model prediction and classification". During the training and testing phase. Observed techniques in these competitions are diverse, including Tree-Based Methods [22, 102, 43], CNNs [69, 24, 62], RNNs and their variants [7, 62, 48, 107], Transformers [111, 124], DANN [49], SVMs [75, 80], Unsupervised Learning, Ensemble Learning [86, 107], Similarity-Based Methods [73, 47, 2], Rule-Based Methods [3, 61], Transfer Learning [66, 120, 8], Domain Adaptation [80, 49], etc. Once the ML models are trained and tested well, they become invaluable tools for various PHM applications which include detecting abnormal performance or faults, classifying different faults, assessing the current health state of systems, or predicting the RUL of components. These applications are not only critical in meeting the diverse requirements of PHM data challenge competitions but also demonstrate the adaptability and effectiveness of the various ML and DL approaches utilized.

Model Interpretability: Except for the pursuit of accuracy in prediction and classification problems, model interpretability is becoming more and more important. Model interpretability refers to interpreting ML model outputs [76] which helps humans understand the 'why' behind a model's predictions, facilitating collaboration and more informed decision making. Many explainable artificial intelligence (XAI) methods such as Local Interpretable Model-Agnostic Explanations (LIME) [88] and SHapley Additive exPlanations (SHAP) [70] are utilized in the PHM data challenge competitions to explain model outputs in PHM [98].

In Section 3 and Section 4, our analysis will categorize and delve deeper into the dual themes of "Detection & Diagnosis" and "Assessment & Prognosis".

3 Detection & Diagnosis

Fault detection and diagnosis are essential components of PHM, tasked with the identification of anomalies and determining failure modes in a timely manner, thereby playing a pivotal role in enhancing system safety and reliability, guaranteeing prompt maintenance interventions, and reducing downtime costs. In this section, we provide a comprehensive overview of the detection and diagnosis challenges encountered in the competitions, as illustrated in Table 3. We sequentially introduce the fundamental competition information, including background, objectives, challenges, and datasets, highlighting the innovative approaches to fault detection and diagnosis problems. A summarization of the competition problems and their datasets is encapsulated in Table 2. Furthermore, we provided a comprehensive summary of the methodologies utilized for detection

Table 2. Overview of Detection and Diagnosis Problems in PHM Data Challenge Competitions

	2021 PHM EU	2022 PHM EU	2022 PHM NA	2023 PHM AP	2023 IEEE
System	Manufacturing Production Line	Printed Circuit Boards	Rock Drill	Spacecraft Propulsion System	Gearbox
Failure Mode	8	1+1+2	10	3	4
Sensor Number	50	NA	3	7	1
Asset	NA	NA	8	4	NA
Operating Condition	2	1+1+1	1	1	2
Data Type	Time Series	Tabular	Pressure Signals	Pressure Signals	Vibration Signals
Volume of Data	Limited (70 Train, 29 Test)	Medium (SPI:1924, AOI:1924)	Large (37229 Train, 16396 Test)	Limited (177 Train, 46 Test)	Large (Total 50000)
Sampling Rate	0.1 Hz	NA	50 kHz	1 kHz	10 kHz
Sampling Interval	1-3 hours	NA	NA	1.2 s	5 minutes

and diagnosis within the PHM data challenge competition at the end of this section, as outlined in Table 4.

3.1 Challenges

Table 3. Overview of Detection and Diagnosis Challenges in PHM Data Challenge Competitions

	Unique Challenges
2021 PHM EU	Missing Data & Noise Data Imbalance Limited Data
2022 PHM EU	Missing Data Data Imbalance Component-level Prediction
2022 PHM NA	Domain Shift
2023 PHM AP	Unknown Fault Limited Data
2023 IEEE	NA
Common Issues: Multiple Fault Modes	

Table 3 provides a detailed summary of the detection and diagnosis challenges specific to each competition. While each competition faces unique challenges due to its distinct industry background and objectives, a common issue they all share is the need to accurately classify multiple fault modes. Apart from the common issue, notably, the PHM EU competition in 2021 and the PHM AP in 2023 are both constrained by their small datasets, with the former comprising only 70 training and 29 test samples, and the latter having 177 training and 46 test samples. This paucity of data presents substantial obstacles for training ML models due to limited information from data. Additionally, in the 2021 and 2022 PHM EU competitions, the datasets show a significant imbalance with a higher number of healthy experiments compared to the faulty ones, thus impeding the models' ability to detect rare fault conditions and challenging their robustness and generalization potential to some degree. In the meantime, these datasets also faced the obstacle of missing data, necessitating imputation techniques for robust data handling. The 2022 PHM NA competition confronts a domain-shift problem, where varying operational conditions and data from different assets result in inconsistent signal distributions, which complicates the task of accurate fault classification. Lastly, the 2023 PHM AP competition includes unexpected fault modes in the testing data, absent from the training set, reflecting the unpredictability of real-world industrial settings and underscoring the necessity for constructing robust and generalized models.

3.2 2021 PHM EU (Manufacturing Production Line)

In a collaborative effort with the Swiss Centre for Electronics and Microtechnology (CSEM), a data challenge competition was introduced, offering a dataset derived from a real-world industrial manufacturing line dedicated to testing electrical fuses. The objective of this competition is to perform fault identification and classification, root cause analysis, rapid prediction, and system operation parameter identification. The dataset showed eight unique system failure modes, replicated under two distinct operating conditions. However, its scale was rather restricted, comprising merely 70 training and 29 testing samples. The dominant challenge with the training data was its class imbalance—most samples were in healthy condition. This scenario, coupled with the limited volume of data, presented challenges to the data-driven approach.

Against this background, the winning solution was a combination of decision tree algorithms and a propagation system [22]. While the decision trees focused on observational diagnoses, the propagation system skillfully tackled chronological intricacies by infusing a Kalman-style filter. This fusion bolstered the reliability of the outcomes. To address data imbalance, the SMOTE (Synthetic Minority Oversampling Technique) method was utilized. Additionally, [41] incorporated the Leave One Feature Out Importance (LOFO-Importance) package for capturing essential features. Subsequently, they applied linear discriminant analysis (LDA) for dimensionality reduction. During the modeling phase, different ML methods were employed — gradient boosting algorithms such as Extreme Gradient Boosting (XGBoost) and Light Gradient Boosting Machine (LightGBM), LDA classifier, and Gaussian process classifier. Especially for gradient boosting, they used Genetic Algorithms to optimize hyperparameters. A pivotal observation was XGBoost's superior performance over the other algorithms. Differing from their approach, [3] showcased a rule-based diagnostic technique, comparing its performance with that of decision trees and random forest methods. Additionally, [7] proposed a regularized LSTM for sifting through vital features and then leveraged an ensemble of binary LSTM classifiers for fault detection and classification.

Moreover, the tree-based algorithm XGBoost found favor not just in the aforementioned studies but also in three other distinct research papers [4, 85, 106] outside the competition. The distinctions among these studies were as follows: For instance, [106] relied on a feature importance ranking (FIR) method, targeting enhanced performance and simplification in intricate industrial classification scenarios, whereas [85] focused on the challenge of missing value imputation, harnessing Partial Least Squares (PLS-MV). A significant contribution of Ramezani's framework was its explainability, achieved by pinpointing key performance indicators for each fault family using the SHAP method [70]. In conclusion, tree-based methods can significantly enhance fault diagnosis, thereby boosting the efficiency and safety of manufacturing production lines.

3.3 2022 PHM EU (Printed Circuit Boards)

The 2022 European PHME Data Challenge, hosted in collaboration with Bitron Spa, spotlights a classification issue within an actual industrial Printed Circuit Board (PCB) production line. The challenge's objectives span three pivotal sub-tasks: predicting Automatic Optical Inspection (AOI) defect detection based on Solder Paste Inspection (SPI) data, predicting human-made visual inspection labels (OperatorLabel), and finally, predicting the human-assigned repair label (RepairLabel). Task 1 and Task 2 are binary classification problems, whereas Task 3 is a multi-class classification problem in this data challenge. This explains why, in Table 2, the "Failure Mode" is listed as "1+1+2". In Table 2, the "Operating Condition" is listed as "1+1+1", which means each task has its own operating condition. It's noteworthy that, among the nine datasets discussed in this paper, this dataset stands unique. While others are time series data, this one is parametric data.

Delving into the competition's outcomes, Gaffet's team clinched 1st place with an XGBoost method based on encoding and feature engineering [32]. Notably, they further harnessed the SHAP method for model interpretation. Similarly, [102] leveraged two tree-based gradient boosting algorithms, namely LightGBM and XGBoost. Their approach centered on solving classification problems by extracting significant statistical data during the feature engineering phase. The importance of feature engineering was further emphasized by [104]. They introduced a novel statistical feature extraction method coupled with a PinNumber-based technique. This method aimed to compress pin-level data into component-level information. When predicting automatic inspection defects, they integrated a neural network model, factoring in feeding imbalance control to navigate data imbalance challenges. Additionally, a random forest model was developed for both human inspection and repair predictions. Similarly, apart from an extensive focus on feature engineering, [89] applied a multi-layer perceptron (MLP) neural network for defect predictions in automated inspections. For human inspection outcomes, a random forest algorithm was their choice, while decision trees were favored for predicting human repair labels.

Outside of competition, Mirzaei's team delved deep into the challenge of imbalanced data [75]. Their solution entailed a data-level technique that integrated recursive feature elimination (RFE) for feature selection and oversampling methods. This blend ensured balanced representation for minority classes, enhancing the performance of multiple ML algorithms, from decision trees and random forests to SVMs and 1dCNNs. In a somewhat divergent approach, [58] introduced a new quality management paradigm termed Stream-of-Quality (SoQ) tailored for multi-stage manufacturing processes. By leveraging this dataset, they showcased the effectiveness of their methodology. The implications of their work are vast, offering promising avenues for refining industrial AI algorithms and methodologies in a systematic manner.

3.4 2022 PHM NA (Rock Drill)

Rock drills play a pivotal role in sectors like mining, tunneling, and construction. Due to the potential economic and human costs from work interruptions caused by rock drill faults, ensuring accurate fault diagnosis is imperative. This data challenge tackles the task of fault classification within various rock drill configurations. Challenges arise due to the domain-shift phenomenon, stemming from data sourced from diverse rock drill machines, which results in heterogeneous signal distributions that potentially impair classification accuracy. The dataset [44] encompasses ten distinct fault modes and one health mode, with training, validation, and testing data sizes of 34,045, 3,184, and 16,396, respectively.

In the competition, the champion was clinched by [80], who deployed a hybrid strategy, combining data-driven techniques with various signal-processing methods. Their approach harnessed domain adaptation, metric learning, and pseudo-label-based deep learning to construct an ensemble DL model for comprehensive fault classification. For samples that posed challenges for DL, they deployed signal processing methods like Dynamic Time Warping (DTW), and Cross-correlation, and used SVM for supervised learning. The runner-up solution introduced a data-cropping technique, employing a convolutional neural network (CNN) as a pivotal feature extractor to bridge data length discrepancies. [49] innovatively addressed the domain-shift issue through a domain-adaptation-based scheme that harnessed a domain adversarial learning neural network for extracting domain invariant features while using maximum mean discrepancy (MMD) minimization for bridging distribution discrepancy, and a soft voting ensemble to reduce model uncertainty. Moreover, Minami's team proposed an ensemble approach, appending specialized models onto a baseline model. This ensemble incorporated domain adaptation strategies to accommodate domain fluctuations. The baseline model employed conventional ML algorithms like SVM, Random Forest, and XGBoost for whole multi-class classification, whereas the specialized sub-models, leveraging CNN for feature extraction and classification, concentrated on binary classifications targeting specific classes that poorly performed on the baseline model [74].

Outside the competition, two notable data-driven strategies emerged. [65] deployed an end-to-end fault classification framework derived from X-Vectors [95], achieving integrated optimization of both feature extraction and classification phases [65]. [103] proposed a novel deep forest algorithm that fused multi-grained scanning for feature extraction and a cascade forest for layered predictions to perform failure model classification. In summary, all aforementioned methods emphasize the importance of tackling the domain shift challenge. Utilizing domain adaptation and ML techniques can enhance the accuracy of fault detection and diagnosis in rock drills, ensuring safer and more reliable operations.

3.5 2023 PHM AP (Spacecraft Propulsion System)

The Japan Aerospace Exploration Agency (JAXA) initiated a competition centered on the advancement of PHM technology for spacecraft propulsion systems. The primary objective was to accurately diagnose various states ranging from normal conditions to bubble anomalies, solenoid valve faults, and unknown faults. Simulation data was sourced from four distinct spacecraft, resulting in a dataset comprising 177 training samples and 46 test samples [108]. The paucity of data posed a significant challenge for the data-driven methods.

In the competition, the champions [73], introduced a novel two-step approach. Initially, a similarity measures-based model was proposed for the categorization of data into four distinct states. Subsequently, for data corresponding to the solenoid valve fault, a model incorporating physics-inspired features was employed to pinpoint the fault location and estimate the valve opening ratio. A noteworthy aspect of their methodology entailed the employment of DTW [11] on the training dataset. This step was instrumental in quantifying the variability across various segments of the sensor data. Based on data analysis, authors opted to leverage solely the initial valve opening data segment for feature extraction, thereby mitigating variations arising from individual or sample variations. This strategic refinement substantially improved the data quality, resulting in better model performance in fault detection and classification, with the team achieving 100% classification accuracy in the final competition. Standing in the second position, [61] devised a hybrid approach combining the XGBoost-based method and the rule-based method. While the XGBoost-based approach primarily addressed comprehensive fault classification, the rule-based method was employed to formulate the solenoid valve opening ratio equation. This was accomplished through polynomial fitting, rooted in intrinsic physical characteristics, enabling a precise estimation of the solenoid valve opening ratio. Additionally, [47] utilized the K-NN algorithms to classify faults and pinpoint the location of anomalies. Prior to the classification, a differentiation between normal and anomalous data was executed based on a similarity-based approach. Their estimation metrics for valve opening ratio were hinged upon the similarity of time series waveforms. Moreover, [2] employed an ensemble framework, integrating K-means clustering and decision trees. This model, enriched by domain-specific expertise, exhibited good preci-

Table 4. Overview of Detection and Diagnosis Methodologies in PHM Data Challenge Competitions

	Methodology		
	Deep Learning	**Conventional Machine Learning**	**Feature Engineering**
2021 PHM EU	Regularized LSTM Ensemble LSTM	XGBoost, Decision Trees Random Forest LDA, Rule-based Method Gaussian Process	FIR PLS-MV FCM SMOTE
2022 PHM EU	1D-CNN	Decision Trees, Random Forest XGBoost, LightGBM SVM, MLP	RFE Statistical Feature Extraction
2022 PHM NA	Domain Adaption DANN, X-Vectors Metric Learning Pseudo Label Technique	Ensemble Learning XGBoost SVM Deep Forest Algorithm	MMD RFE DTW
2023 PHM AP	NA	Similarity-based Method K-means clustering, KNN Decision Trees, XGBoost Rule-based method Ensemble Learning	Physical Feature Extraction DTW
2023 IEEE	ROCKET, LSTM-FCN 1D-CNN with ResNet Deep Residual Network Residual-based CNN	Ensemble Learning	Data Augmentation Data Regularization STFT

sion in both anomaly detection and fault diagnosis. Various approaches in the competition underscore the pivotal role of similarity-based methods and the extraction of physical features. These approaches stand out as potent tools in the realm of fault detection and classification, especially when contending with limited datasets.

3.6 2023 IEEE (Gearbox)

Gearbox malfunctions in industrial machinery can result in significant downtime and damage, leading to substantial financial losses. Early detection and accurate diagnosis of these faults are paramount for ensuring the machinery's safe operation, averting catastrophic failures, and minimizing maintenance costs. The objective of this competition is to develop ML-based models that can efficiently detect faults in the planetary gearboxes of industrial machinery using vibration signals. The dataset [29] spotlights four prevalent sun gear faults: surface wear, chipped teeth, cracks, and missing teeth. Vibration signals have been recorded for a duration of five minutes each, with a sampling rate of 10 kHz, under two distinct operational conditions. Given the expansive data size, totaling 50,000 entries, the potential for deploying DL techniques is evident.

Various data-driven methodologies have been employed in this competition. For instance, [62] utilized ensemble-based CNN techniques to address fault detection via time-series vibration data. Their findings underscored the robust performance of integrating three convolution kernel-based methods such as ROCKET (RandOm Convolutional KErnel Transform) method [23], one-dimensional convolutional neural networks integrated with ResNet, and a fusion of LSTM and Fully Convolutional Network (FCN) in delivering classification results for multivariate time-series data. Similarly, both [91] and [51], designed residual-based CNN models to address fault classification challenges. On the one hand, [91] incorporated the Short-Time Fourier Transform (STFT) [40], transforming frequency domain signals into time-frequency domain signals using Fourier analysis - an invaluable tool for interpreting time-evolving signals. On the other hand, [51] leveraged data augmentation and regularization techniques, enabling model construction with fewer parameters without sacrificing performance quality. In another noteworthy contribution, [96] harnessed a tree classifier to select valuable features from raw vibration signals, subsequently developing a sequential neural network model tailored for the concurrent detection of multiple gear faults.

4 Assessment & Prognosis

This section delves into the health assessment and prognosis problems within PHM, emphasizing their role in evaluating the current health state of industrial machinery and estimating their RUL. Assessment and prognosis are pivotal to implementing predictive maintenance strategies and preemptively addressing system degradation, thereby reducing the incidence of unplanned downtime, extending equipment lifespan, and optimizing maintenance schedules. Similar to the structure of Section 3, we first summarize the challenges in the competition as outlined in Table 5. Subsequently, we detail the specifics of each competition, including the data-driven approaches applied, and provide a consolidated summary of both the problems with their associated datasets and the proposed solutions in Table 6 and Table 7, respectively.

4.1 Challenges

The pivotal challenge across all competitions in health assessment and prognosis is summarized in Table 5. The common challenge focuses on identifying degradation trends and estimating RUL. In the meantime, diverse industrial scenarios present each competition with unique challenges; for instance, the 2018 PHM NA and 2020 PHM EU are both confronted with variability due to differing operational conditions—sensor discrepancies due to varied tool use in the former, and twelve distinct operational conditions influenced by solid ratios and particle sizes in the latter. Similarly, in 2021 PHM NA, the diverse flight envelopes introduce considerable variability into the data. A common thread in the 2018 and 2021 PHM NA competitions is the critical need for models to robustly classify multiple fault modes to enable precise RUL prediction. Complicating this need is the data imbalance evident in the 2018 PHM NA, where rare fault instances are overshadowed by prevalent healthy data, potentially reducing the models' capabilities for rare fault detection. In the 2019 PHM NA, the limited availability of ultrasonic signal data, particularly for the validation set, poses a risk of model overfitting, potentially reducing the reliability of data-driven prognostics. The 2020 PHM EU is faced with the uncertainty of unknown faults, as the test set includes operational conditions not represented in the training and validation data, a discrepancy that could lead to diminished model performance. These multifaceted challenges necessitate the development of sophisticated, adaptable ML models capable of overcoming data discrepancy, data imbalance, and operational condition variability to deliver accurate and reliable health assessments and prognostic predictions.

Table 5. Overview of Assessment and Prognosis Challenges in PHM Data Challenge Competitions

	Unique Challenges
2018 PHM NA	Data Discrepancy & Data Imbalance Different Operating Conditions Multiple Fault Modes
2019 PHM NA	Variable Loading Conditions Limited Physical Information Limited Data
2020 PHM EU	Different Operation Conditions Unknown Fault
2021 PHM NA	Multiple Fault Modes High Variability in Flight Envelopes
Common Issues: Degradation Trend Detection	

4.2 2018 PHM NA (Ion Mill Etching System)

In the quest to advance predictive maintenance strategies for ion mill etching systems within semiconductor manufacturing, the diagnosis and prognosis of three principal failure modes—Flowcool leak (F1), Flowcool Pressure Too High Check Flowcool Pump (F2), and Flowcool Pressure Dropped Below Limit (F3)—are critical. Successful prediction of these failures helps in the timely scheduling of maintenance operations, thereby reducing downtimes and enhancing operational efficiency. The 2018 PHM North America data challenge emphasized the analysis of fault behavior within the ion mill etch tool, tasking participants to develop a model from the sensor-derived time series data capable of accurately detecting, diagnosing, and prognosticating the time-to-failure for these malfunctions.

Investigation into the proposed solutions reveals a pronounced inclination towards tree-based methodologies for fault detection and diagnosis. [118, 38, 124] and the team led by [34] have each proposed strategies based on random forest algorithms for early degradation mode detection and diagnosis. Extending the exploration of ML techniques, [93] evaluated an assortment of models, including Logistic Regression, Generalized Linear Models, Multivariate Adaptive Regression Splines, Support Vector Regression (SVR), MLP, random forest, etc. Among these, the random forest model distinguished itself with superior performance. Meanwhile, [126] utilized a knowledge distilla-

Table 6. Overview of Assessment and Prognosis Problems in PHM Data Challenge Competitions

	2018 PHM NA	2019 PHM NA	2020 PHM EU	2021 PHM NA
System	Ion Mill Etching System	Fatigue Crack	Filtration System	Turbofan Engine
Failure Mode	3	1	1	7
Sensor Number	5	2	3	14
Asset	20	8	NA	NA
Operating Condition	Multiple	Variable Loading Conditions	12	4
Data Type	Time Series	Time Series	Time Series	Time Series
Volume of Data	Limited Samples (20 Train, 5 Test)	Limited Samples (74 Train, 36 Test)	Limited Samples (24 Train, 8 Validation, 16 Test)	Limited Samples (90 Train Units, 38 Test Units)
Sampling Rate	0.25 Hz	5 Hz	10 Hz	1 Hz
Sampling Interval	Above 70 million seconds	14000-100774 cycles	200-350 s	1-3 hours, 3-5 hours, Above 5 hours

tion approach towards fault detection across various modes, aiming to bolster the detection of infrequent yet critical faults.

Recent advancements in RUL prediction [48] have predominantly hinged on the application of DL algorithms to analyze complex multivariate time series data. [118, 38, 34] have all utilized Long Short-Term Memory (LSTM) networks and their variations, such as Gated Recurrent Units (GRU) and LSTM-based Metric Regression (LSTM-MR), to capture important features from the raw data. [35] expanded upon this approach by integrating a Temporal Convolutional Network (TCN) with LSTM, enhanced by an attention mechanism, which facilitated refined feature extraction from sensor data for accurate RUL prediction. What's more, Liu et al.'s two-stage deep transfer learning framework aimed at achieving accurate RUL prediction. In the first stage, the developed model leveraged TCN for initial temporal feature learning, followed by domain adversarial learning for data alignment based on one fault mode. Then in the second stage, the first-stage model was fine-tuned based on other fault mode data to handle multiple fault modes and enhance the RUL prediction performance [66]. Distinct from these methods, [124] explored transformer networks, focusing solely on data from abnormal operation phases, while [68] provided a comprehensive comparison of state-of-the-art methods, including TCN, LSTM, attention-based mechanisms, CNN, and Transformer, culminating in the development of the CeRULEo library to facilitate research in the field. Moreover, [93] introduced a novel approach using DTW for RUL estimation, leveraging a library of truncated degradation curves and health score models to refine the final RUL predictions. Such diversity in methodologies not only underscores the complexity of fault diagnosis and prognosis in high-tech manufacturing but also showcases the robustness of ML in tackling these challenges.

4.3 2019 PHM NA (Fatigue Crack)

The 2019 PHM Conference Data Challenge put a spotlight on the task of fatigue crack length estimation and prediction within aluminum structures at different points. The challenge harnessed wave signal data collected by piezoelectric sensors subjected to both static and dynamic tensile stresses. Given that the test dataset was limited to wave signals from several initial loading cycles, participants were challenged to estimate crack lengths where signal data were present and predict future crack growth for specified cycles lacking signal data. The complexity of the problem delineated a clear path: data-driven methodologies were viable when signals existed, whereas scenarios lacking wave data required exploration into physics-based strategies. Reflecting this path, solutions derived from this challenge have predominantly employed a hybrid approach of data-driven and physics-based techniques.

Regarding the estimation of fatigue crack length with accessible wave signals, a variety of data-driven models and feature engineering strategies emerged, both within and beyond the competition's scope. [46] proposed a neural network architecture reliant on features manually derived from raw signals, such as the Pearson correlation coefficient, phase shifts, energy, and information entropy, to train models for accurate crack length estimation. Similarly, [50] initiated their approach with signal preprocessing, utilizing techniques like band-pass filtering and phase alignment to mitigate noise and uncertainty before applying physically insightful feature extraction methods. Thereafter, they harnessed a random forest algorithm, optimizing it through feature selection and grid search for hyperparameter finetuning, to estimate crack lengths. Consistent with these methods, [121] also leveraged a band-pass filter for feature extraction from raw wave signals, subsequently constructing an SVR model with hyperparameters optimized via grid search. Rao and collaborators, meanwhile, designed an ensemble learning regression model to improve estimation performance with four useful extracted features (root mean square value, correlation coefficient, first peak value, and the logarithm of kurtosis) [86].

Predictive modeling for scenarios lacking wave signal data required a pivot towards physics-based techniques, often in conjunction with data-driven insights, to forecast crack progression. [46] devised a Particle Filter (PF) strategy, integrating the Paris Law and outputs from the previously developed neural network as observational inputs to refine and update the crack propagation pathway. [50] suggested an ensemble prognostics framework under consistent loading conditions, constructing a probability density function (PDF) for each instance. This was followed by a computation of weights derived from each PDF to output the final crack length prediction. Furthermore, when different loading conditions prevailed, Walker's equation was utilized to forecast crack lengths. Innovatively, [121] introduced a trans-fitting approach, aimed at extracting the crack growth trend from training data and extrapolating it to the test data predictions. Additionally, Rao's group advanced a variation version of Paris' Law, aiming to elucidate the correlation between crack progression and the number of loading cycles [86].

4.4 2020 PHM EU (Filtration System)

In the domain of industrial maintenance, filtration systems are pivotal for helping process pollutants from industrial equipment, ensuring seamless system operation. A predominant challenge encountered within these systems is filter clogging - a phenomenon where accumulated pollutants impede flow rates, thereby disrupting standard industrial workflows. Addressing this issue, the PHM-Europe Conference 2020 data challenge concentrated on the prediction of filtration systems' RUL, crucial for timely maintenance and replacement scheduling. RUL, in this context, is delineated as the time until the pressure differential across the filter breaches a threshold of 20 psi. The challenge provided the PHME20 dataset, collected from a controlled experimental setup designed to simulate filter clogging at various contamination levels. Twelve distinct conditions were established influenced by two operational parameters: solid ratio (%) and particle size (µm).

The champion solution [67] employed a novel hybrid approach, combining kernel regression with fundamental statistical methodologies. This approach, underscored by complex data analysis, proved efficacious in enhancing the RUL prediction. Meanwhile, the runner-up, [10], adopted different cutting-edge feature engineering, and ML methods. Their process began with the extraction of features through a rolling window technique and proceeded with feature selection informed by the linear kernel Support Vector Machine (SVM) coefficients, RFE, correlation matrix, and monotonicity testing. A four-layer sequential neural network was their model of choice, supported by K-fold cross-validation throughout the training and validation stages. Capturing the third position, [43] conducted a comparative study of tree-based algorithms and the Bayesian approach. Specifically, random forest, gradient boosting, and Gaussian process regression were utilized to estimate the RUL of the filtration system, supplemented by a novel fault-based RUL assignment that integrated "Piecewise RUL Assignment" and "Linear RUL Assignment".

Contrasting with the feature engineering and conventional ML strategies, some researchers explored some DL methods. [113] introduced a CNN-based DL methodology for RUL prediction, marked by two architectural innovations: a Parameterized Fully Connected Layer that adjusts network weights in response to operational parameter shifts, and a multi-head predictor tailored to distinct degradation process stages. Additionally, [107] presented a new transfer ensemble learning (TEL) framework, leveraging metric learning alongside domain dissimilarity metric and Kullback–Leibler (KL) divergence, to enhance model generalization from source to target domains. This TEL framework amalgamated with a bidirectional long short-term memory (Bi-LSTM) algorithm, coined as TEL-Bi-LSTM, was offered for RUL estimation under different operating conditions. In another innovative approach, [42] proposed a joint autoencoder-regression network, a deep neural architecture that fused a CNN autoencoder with an LSTM network regressor in an end-to-end training paradigm. Genetic algorithms were instrumental in optimizing hyperparameters for this architecture. Additionally, Lee's team developed a distinct strategy by first establishing a health assessment criterion [60]. They defined a Health Index (HI) for the filter system and utilized K-means clustering for the categorization of the system's health stages. Subsequent HI predictions were facilitated by the Bi-LSTM algorithm, thus determining the system's RUL. Lastly, Falcon et al. introduced an innovative sequence modeling technique termed the Neural Turing Machine (NTM) [30]. Conceptualized as a computational architecture, the NTM leverages available data to interact with an external memory component, an approach that facilitates enhanced accuracy in predictions. This model stands out for its ability to generate more precise outcomes when benchmarked against the prevalent LSTM-based solutions that dominate the field.

4.5 2021 PHM NA (Turbofan Engine)

The 2021 North America PHM Data Challenge was primarily focused on the prediction of RUL for turbofan en-

Table 7. Overview of Assessment and Prognosis Methodologies in PHM Data Challenge Competitions

	Methodology		
	Deep Learning	**Conventional Machine Learning**	**Feature Engineering**
2018 PHM NA	Transfer Learning LSTM, LSTM-MR, GRU Transformer CNN TCN, TCN-LSTM, TCN-DANN	Random Forest, Gradient Boosting Logistic Regression Generalized Linear Model SVR, MLP MASR	DTW SVR-RFE Degradation Curve Library
2019 PHM NA	NA	Random Forest, Gaussian Process SVR Neural Network Ensemble Learning Linear Regression	Grid Search Genetic Algorithm Paris's Law Walker's Equation Particle Filter
2020 PHM EU	Neural Turing Machine Transfer Ensemble Learning LSTM, Bi-LSTM, TEL-Bi-LSTM Autoencoder-Regression Network Deep CNN	Random Forest, Gradient Boosting Gaussian Process Kernel Regression SVR Ensemble Learning	Simple Statistics Method Linear SVM Coefficient Correlation Metric RFE, Health Index Monotonicity Test
2021 PHM NA	Deep CNN, FCN, VGG ResNet(Residual Block) GoogLeNet(Inception Module)	Random Forest, XGBoost Extreme Random Forest ANN	PCA XAI,SHAP, LIME

gines [13], specifically under four distinct flight conditions and seven failure modes. Participants were required to create predictive models leveraging the N-CMAPSS dataset, aiming to accurately forecast RUL using complex condition monitoring data. The dataset, a collection consisting of 90 synthetic run-to-failure trajectories for training and an additional 38 truncated datasets for testing, served as a comprehensive foundation for developing robust predictive algorithms to predict RUL accurately.

During the competition, the winning team, led by Lovberg, put forward an innovative approach leveraging a deep convolutional neural network (DCNN) [69]. This network was distinguished by its use of dilated convolutions complemented by gated linear unit activations and integration of residual skip connections. Those techniques were designed to expand the network's receptive field and enhance flexibility, so as to reduce the complexity of the neural network architecture by using less number of parameters, but still having comparable performance. Additionally, they adopted a strategic sequence sampling method, minimizing less informative samples while retaining enough degradation signals for the network's input. This methodology distinguished their solution and got 1st place in the PHM data challenge competition. Moreover, Solis-Martin et al. and DeVol's team pursued advancements in DCNNs as well. The former developed a two-level DCNN system where the first-level DCNN focused on extracting vital features from raw data and the second-level DCNN leverages the output from the first level to accurately estimate the RUL [97]. The latter drew upon established DL architectures, deploying the basic blocks or modules from the VGG, GoogLeNet, and ResNet designs into their DCNN framework. This enabled a comparative analysis of model performances using a variety of well-known architectural features [24, 25].

Outside of the competition, [19] addressed concerns regarding the uncertain and poor interpretability of deep learning models by integrating principal component analysis (PCA) to refine time-domain feature sets and subsequently applying four supervised learning techniques, including artificial neural network, random forest, extreme random forest and XGBoost, to estimate RUL. Their innovative use of a custom loss function in conjunction with traditional ANN models got the best results in both Area Under the Receiver Operating Characteristic (AUROC) and Area Under the Precision-Recall (AUPR) metrics. Moreover, the domain of PHM has witnessed an upsurge in the application of XAI techniques to bolster the interpretability and trustworthiness of ML models. Various methods, including LIME, SHAP, LRP, Image-Specific Class Saliency Maps, and Gradient weighted Class Activation Mapping (Grad-CAM), have been reported in recent literature [18, 20, 98]. These techniques are increasingly critical in elucidating the decision-making processes of complex models, thereby fostering greater confidence in their predictive capabilities.

5 Challenges and Possible Solutions

By reviewing nine open PHM data challenge competitions over the past six years, these competitions have showcased a variety of ML approaches applied to different industrial systems. In this section, we summarize common challenges regarding data-related issues and model-related issues and analyze relevant solutions applied to solve these challenges. Moreover, we evaluate the limitations of these competitions and suggest future directions for their evolution, aiming to accelerate the development of next-generation ML-driven PHM solutions.

Data-related challenges encompass issues such as missing data, data imbalance, and domain shift while model-related challenges include the critical aspects of model selection, interpretability of ML models, and their robustness and generalization capabilities. These challenges highlight the complexities of developing effective ML methods to solve various PHM problems. The limitations of current PHM competitions reveal a gap in adopting systematic approaches for building effective PHM systems, and a need for multi-modal machine learning analysis.

5.1 Data-related Issues

5.1.1 Missing Data

Handling missing data in ML, especially in the context of the PHM is significant, as incomplete data can significantly impact the model performance and prediction accuracy. The strategies to address missing data in PHM have evolved, encompassing a range of techniques from basic deletion to advanced imputation methods. This was evident in the 2021/2022 PHM EU competitions, where missing data posed significant challenges. In the fuse test bench dataset (2021 PHM EU), about 10% of the data was missing, and it was not evenly distributed across variables. Meanwhile, the SPI dataset(2022 PHM EU) had missing information in specific fields. To tackle these issues, various approaches were employed.

A straightforward strategy is the deletion of data points with missing values, such as listwise or pairwise deletion. For example, in the 2022 SPI dataset, instances having missing values in crucial identifiers like "Panel_ID", "Figure_ID", and "Component_ID" were eliminated [102]. However, this approach can lead to the loss of valuable information to some degree. On the imputation front, several innovative methods were used in the 2021 PHM EU competition. One team applied the Last Observation Carried Forward (LOCF) method, coupled with backward filling, to address the gaps [22]. Interpolation was another technique used to estimate missing values, ensuring that cases with absent data did not skew the results. Additionally, Ramezani's team introduced a novel approach, PLS-MV, a partial least squares-based method for imputing missing values [85]. In addition to various imputation methods highlighted in the competitions, there remains scope for exploration in future research such as mean/median/multiple, K-Nearest Neighbors(KNN), regression imputation, maximum likelihood estimation, and ML-based approaches [39]. These methods offer a promising avenue for further enhancing data quality and model reliability in PHM systems.

5.1.2 Data Imbalance

Data imbalance arises when the distribution of classes in a dataset is uneven. This issue is especially pronounced in PHM due to the rarity of failure events in comparison to data representing normal operations. Data imbalance can significantly undermine the performance of ML models, particularly in classification tasks. When trained on imbalanced data, models may become biased towards the majority class (normal operation) and may not effectively recognize the minority class (failure). This leads to poor performance in predicting failures, which may bring ineffective maintenance planning and unexpected downtimes. In PHM data challenge competitions, this issue is commonly encountered. Fuse test bench dataset from the 2021 PHM EU competition and printed circuit board dataset from the 2022 PHM EU competition exemplify this issue. The former displayed an obvious imbalance with only 3-4 experiments representing each fault class against 70 experiments for the healthy class. Similarly, in the printed circuit board dataset, 95% of the SPI data was classified as healthy, highlighting a significant imbalance. Additionally, the 2018 PHM NA competition also faced this challenge in the Ion Mill Etching (IME) dataset, where the faulty data is much less than normal operation data.

To counteract data imbalance in PHM, various strategies have been adopted. Resampling techniques are commonly used in competitions to adjust the dataset to balance the class distribution, either by oversampling the minority class or downsampling the majority class. Synthetic data generation is another approach, as demonstrated by a team in the 2021 PHM EU competition that used the SMOTE to create synthetic samples of the minority class to balance the dataset [22]. Advanced algorithms also play a crucial role in addressing data imbalance. For instance, Tang's team in the 2022 PHM EU competition used the FIR to control the imbalance ratio in each mini-batch during neural network training [104]. Liu et al. employed transfer learning and domain adaptation techniques in the 2018 PHM NA competition to facilitate knowledge transfer across different fault modes, addressing the issue of insufficient data in specific faults [66]. Moreover, during the 2018 PHM NA competition, many teams utilized Random Forest, an ensemble learning method known for its proficiency in handling imbalanced data, by constructing a forest of decision trees. These approaches collectively demonstrate the diverse and innovative methods employed to mitigate the challenges posed by data imbalance in the PHM field.

Beyond the techniques showcased in the competitions, new DL approaches are emerging to address this issue, such as the semi-supervised information maximizing generative adversarial network [117], the integration of deep residual networks with auxiliary classifier generative adversarial networks [14], the combination of DL with SMOTE [21], etc. Furthermore, Aguiar et al. introduced a standardized experimental framework to evaluate 24 state-of-the-art data stream algorithms across 515 imbalanced data streams [1]. Going forward, there is significant potential for the exploration of additional DL-based strategies to enhance the handling of data imbalance in the PHM domain.

5.1.3 Domain Shift

Domain shift refers to the changes in the data distribution between the training phase (source domain) and the real-world application phase (target domain) of the ML models. This phenomenon frequently occurs when models, initially trained on data from a specific set of machines or under certain conditions, are subsequently applied to different machines or varied operating conditions. Such a shift can markedly affect the performance and reliability of ML models in PHM, making tackling with domain shift problem essential for maintaining the robustness and effectiveness of PHM systems. Transfer learning has emerged as a primary solution to domain shift challenges, as it can achieve the adaptation of models from one domain to be effective in another through fine-tuning and domain adaptation techniques. Additionally, robust modeling approaches and the ability to continuously update models with new data are also useful and imperative for mitigating the impact of domain shift.

A clear example of domain shift was observed in the rock drill dataset from the 2022 PHM NA competition. The dataset encompassed training data from five rock drill machines, validation data from another machine, and test data from two additional machines, highlighting potential domain shift issues between training, validation, and test data.

Table 8. Summary of Methodologies Solving Data-related Issues in PHM Data Challenge Competitions

Data Issue	Potential Solution
Missing Data	(1) Listwise or Pairwise Deletion (2) Imputation (LOCF, PLS-MV) (3) Other
Data Imbalance	(1) Resampling (Oversampling or Downsampling) (2) Synthetic Data Generation (SMOTE and Variants) (3) Transfer Learning Techniques
Domain Shift	(1) Transfer Learning Techniques (2) DANN

Similarly, the 2020 PHM EU dataset demonstrated domain shift, with training data featuring small and large particle sizes, while the test data included medium particle sizes. To address these challenges, various strategies were employed. In the 2022 PHM NA competition, Kim et al. utilized domain adversarial neural networks (DANN) along with the minimization of MMD to tackle domain discrepancies [49]. Meanwhile, Oh et al. employed the deep CORAL method, calculating coral loss to diminish domain discrepancy effects by aligning the covariance of the six training domains with that of the test domain [80]. For the 2020 PHM EU competition, Tian's team developed a novel TEL framework, facilitating knowledge transfer from the source to the target domain [107]. These approaches illustrate the potential to overcome domain shift problems in PHM, ensuring that models can maintain accurate predictions despite changes in their working or operation conditions.

5.2 Model-related Issues

5.2.1 Model Selection: Conventional Machine Learning or Deep Learning?

In PHM data challenge competitions, participants face the critical decision between employing conventional ML algorithms or opting for advanced DL models. This choice is influenced by various factors: the volume and quality of available data, and the complexity or nature of the research task (whether it be classification or prediction). An analysis of research papers from the past six years reveals insightful trends and preferences in model selection:

(1) The volume and quality of training data play a significant role in determining the choice of modeling approach. Research teams often choose DL for building data-driven models when training data are abundant. Conversely, in scenarios with limited training data, conventional ML methods, augmented by feature engineering, are more commonly employed.

This trend is clear when addressing classification problems within PHM competitions. As depicted in Table 2, Table 5, and Figure 2, 2022 PHM NA and 2023 IEEE competitions, because of their large datasets, have facilitated the application of DL techniques. Specifically, the 2023 IEEE competition saw widespread use of deep CNN models, including ensemble-based CNNs [62] and residual-based CNNs [91, 51], showcasing advanced DL's capability to tackle complex problems. Additionally, 2022 PHM NA competition favored diverse DL approaches like metric learning + pseudo label-based DL [80], CNN + DANN [49], X-Vectors [65], indicating a preference for integrating transfer learning with DL for enhanced performance. However, the scenario differed for the 2022 PHM EU competition, which provided approximately 2000 samples and encountered data imbalance challenges. Given these constraints, many teams leaned towards tree-based methods known for their robustness and ability to address data imbalance, deploying algorithms like XGBoost [32, 102], LightGBM [102], decision trees [75], and random forest [104, 75]. In competitions with significantly fewer data samples, such as the 2021 PHM EU and 2023 PHM AP, where the dataset sizes were around 100-200 samples, the deployment of DL was impractical due to its requirement for a large volume of data. Instead, conventional ML methods proved more effective. For the 2021 PHM EU, tree-based methods were predominantly used [22, 41, 4, 85, 106], while the 2023 PHM AP competition saw a variety of approaches, including similarity-based [73, 47], rule-based [61], KNN/K-means [47, 2], and tree-based methods [61, 2], emphasizing the adaptability of traditional methods to limited data scenarios.

In the realm of RUL prediction tasks, the inherent long-time series nature of the datasets, even when datasets are smaller (less than 100 samples), allows for different approaches to data utilization and data augmentation. By considering each time point or a sequence of time points (data window) as an independent training sample, the effective size of the dataset can be substantially increased. This is evident from Table 7 and Figure 2, which indicate the application of both DL and conventional ML methods across various competitions, with the notable exception of the 2019 PHM NA competition, where only conventional ML methods were employed. As detailed in Figure 2, competitions such as 2018 PHM NA, 2020 PHM EU, and 2021 PHM NA saw widespread adoption of diverse DL techniques, ranging from CNN [113, 97, 24], RNN-based [38, 118, 107] models to transfer learning [66, 107], Autoencoder-based [42], and transformer-based [124, 68] methods, to tackle RUL prediction challenges. Conversely, the 2019 PHM NA competition's unique challenge, due to the partial absence of wave data in the test dataset, necessitated a hybrid approach that integrated data-driven (conventional ML) and physics-based approaches [50, 121] to address the problem.

(2) The complexity or nature of the research task influences the choice of modeling approach to some degree. Figure 3 shows that tree-based methods (such as Decision Trees, Random Forest, and XGBoost) are favored in conventional ML and DL solutions often favor LSTM-based and CNN-based approaches. A closer examination of Figure 2, Table 4, and Table 7 reveals key insights. The choice between conventional ML and DL for detection and diagnosis (classification problem) hinges largely on dataset size, as discussed before. Absent this factor, for the RUL prediction problem, the majority of teams opted for DL approaches. This is because RNN-based methods (LSTM, GRU, etc) are proficient in handling sequential, time-series data, and CNN-based methods are adept at feature extraction from multidimensional datasets, enhancing the capability to tackle RUL prediction problems efficiently. While conventional ML methods have also shown efficacy, the scalability of DL indicates its superior potential for RUL prediction, given sufficient data. When domain shift is present in the competition dataset, transfer learning and domain adaptation methods are often proposed to adjust the model to new data distributions, and address limitations in training data. This adaptability ensures the model's applicability and effectiveness under varied operational conditions, a frequent occurrence in PHM scenarios.

(3) When datasets become publicly accessible, there is a clear shift towards the adoption of more sophisticated DL approaches in PHM data challenge competitions over time. Taking the 2018 PHM NA competition as a case study, our review of 12 published papers utilizing this dataset reveals that the methodologies initially favored by the competition teams largely comprised tree-based methods, SVMs, and basic LSTM algorithms [38, 93]. However, as time goes by, there is a discernible trend towards the development of more complex DL algorithms. This includes but is not limited to, GRU [118], knowledge distillation [126], a two-stage deep transfer learning framework utilizing TCN and DANN [66], TCN combined with LSTM [35], attention mechanisms [35, 68], and transformers [124, 68]. This trend underscores a growing need for advancing DL approaches in the PHM domain, to deepen dataset understanding and enhance model performance. The continuous innovation and refinement of DL methods indicate a vibrant future for PHM, where advanced data-driven analyses pave the way for novel discoveries in the field.

(4) A general discussion on the advantages and limitations of CNN, RNN, and tree-based methods in the

context of PHM reveals: (1) CNNs excel in extracting features from multidimensional data but struggle with interpretability and require substantial computational resources. (2) RNNs are adept at handling sequential time-series data, benefiting from their ability to remember information over long sequences, useful for understanding long-term dependencies. These approaches are helpful for health state monitoring and RUL prediction, though their complexity and data demands can be challenging. (3) Tree-based methods are valued for their interpretability, feature importance analysis, ability to handle various data types, and effectiveness in capturing non-linear relationships, but they might not achieve the neural networks' accuracy on large complex datasets.

In summary, selecting an appropriate ML approach should consider the dataset's specific characteristics, aiming to balance accuracy, interpretability, and computational efficiency to tackle PHM's diverse challenges. When training data are sufficient, DL methods typically offer superior performance compared to conventional ML approaches.

5.2.2 Machine Learning Model Interpretability

In PHM data challenge competitions, accuracy in prediction and classification is still the primary goal. However, with the advance of ML and DL, emphasizing the importance of model interpretability is becoming as crucial as their accuracy in prediction and classification due to its significant impact on safety, operational decision-making, and compliance with industry standards. Model interpretability not only fosters trust and facilitates collaboration between data scientists and domain experts, but it also plays a pivotal role in error analysis and model refinement in PHM.

While only a few teams in competitions have used model interpretability methods to explain the output of their models, outside of competitions, various techniques have been developed and applied to enhance ML model interpretability [64, 6]. As the complexity of deep neural network models, often referred to as "black boxes", some previous research utilized interpretable methods like Layer-wise Relevance Propagation (LRP), Gradient-weighted Class Activation Mapping in CNNs, and attention mechanisms in sequential models to shed light on model decision-making processes [98]. Additionally, model-agnostic methods like LIME [88] and SHAP [70, 101] have been instrumental in offering insights into model behavior. For instance, Baptista et al. applied the SHAP model to evaluate the outcomes of three different algorithms (Linear Regression, MLP, and Echo State Network) using the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset (jet engines) [9]. Moreover, Moradi et al. introduced an interpretable artificial neural network designed for the automatic selection and fusion of features to develop optimal health indicators from data gathered through structural health monitoring (SHM) [77]. From a broader prospect, the challenge in PHM is to balance the trade-off between model accuracy and interpretability. Achieving this balance is not just about enhancing model performance, but also about building trust and ensuring compliance with industry standards. Furthermore, in an era increasingly focused on ethical and responsible AI, transparent and interpretable models are key to not only enhancing the technical aspects of AI solutions in PHM but also extending to ensuring their successful integration and acceptance in real-world applications [112].

5.2.3 Model Robustness and Generalization

In the realm of PHM, model robustness and model generalization are critical factors that determine the effectiveness of a model in real-world applications. Model robustness is defined as the ability of a model to maintain its performance when facing diverse challenges like noise, outliers, and adversarial examples. Techniques such as data augmentation, where training data can be expanded by creating modified versions of existing data or synthesizing new data, have become commonplace. For instance, in the 2023 IEEE competition, Kreuzer's team employed methods like additive white Gaussian noise, circular shift, and random amplitude scaling to increase the volume of training data [51]. Moreover, regularization techniques like L1 and L2 regularization have been instrumental in preventing overfitting and enhancing stability, making the model less sensitive to small fluctuations in input data. What's more, Ensemble methods have been increasingly recognized for their significant contribution to model robustness in PHM. A comprehensive analysis of recent studies shows that out of sixty research papers, nine utilized various ensemble techniques - including decision trees, random forest, XGBoost, ensemble LSTM [7], soft voting ensemble [49], transfer ensemble learning [107], ensemble regression [86], and ensemble CNN-based [62] approaches - across six different PHM data challenge competitions. This trend underscores the effectiveness of ensemble methods in bolstering model robustness, particularly by reducing the influence of noise and outliers, thereby enhancing the reliability of predictive models in PHM applications. Additionally, adversarial training, which involves training models on both regular and adversarial data, has been recognized for its potential to fortify models against adversarial attacks [117, 49, 83].

Alongside robustness, model generalization has also been a focal point in PHM, with the goal being to develop models that perform reliably on new, unseen data. One key technique for improving generalization is cross-validation, which involves dividing the training data into several subsets and each time using one of the subsets as the validation data and others as the training data for model validation during the training process. In a notable example from the 2021 PHM NA competition, researchers opted for k-fold repeated random subsampling validation. This approach was chosen as an alternative to standard k-fold cross-validation to address its limitation, wherein the size of the validation set diminishes as the number of folds (k) increases [97]. In addition to cross-validation, transfer learning and domain adaptation methods have been crucial in maintaining model effectiveness and generalization across varied data distributions. These methods are particularly relevant in scenarios characterized by ”domain shift,” where the data from the target domain differs from the source domain. The specific implementations of these methodologies in PHM competitions have been detailed in the earlier subsubsection titled “Domain Shift”. Outside of PHM data challenges, some novel approaches are proposed to deal with model generalization issues. Matthew Russell and Peng Wang adopted a domain adversarial transfer learning method inspired by generative adversarial networks, utilizing a 1D CNN architecture to predict tool wear on unseen domains using NASA's milling dataset [115]. Ding et al. developed a multi-source domain generalization learning approach (GRU + Transformer) that can effectively learn useful degradation feature representations from various run-to-failure datasets of internal combustion engine journal bearings across different conditions and predict unseen working conditions well [26]. Furthermore, Ding et al. proposed an adversarial out-domain augmentation (AOA) framework for predicting the RUL of bearings under unseen conditions. The effectiveness of this AOA-based RUL prediction was validated using IEEE PHM Challenge 2012 and XJTU-SY run-to-failure datasets, illustrating its robustness in domain generalization for predictive maintenance [27].

5.3 Limitations of Current PHM Competitions and Opportunities

5.3.1 A Lack of Multi-Modal Machine Learning Analysis

Our review of PHM data challenge competitions reveals a reliance on single-modality data in the competitions, such as pressure signals, currents, vibrations, or images, without incorporating multi-modal datasets for fault diagnosis and prognosis. Multi-modal machine learning (MMML) in PHM refers to capturing complementary information from multiple sources (different data types) to achieve a more comprehensive and precise evaluation of PHM tasks [84, 109]. Despite its potential, it is still underexplored in the PHM domain.

Recently, Jiang et al. leveraged two modality data (vibration and current signals) to develop deep belief networks (DBNs) for diagnosing wind turbine gearbox faults [45]. Su et al. proposed an MMML model using parametric specifications, text descriptions, and images of vehicles to predict five vehicle rating scores [100]. Additionally, Fan et al. evaluate several MMML strategies to create a comprehensive PHM system for coolant pumps in commercial heavy-duty vehicles, utilizing data from onboard signals, multi-dimensional histograms, and categorical variables [31]. Wang et al. proposed a novel method for feature fusion in multimodal data (vibration and torque signals), applying it to diagnose the bearings faults [114].

Literature on MMML for PHM tasks shows that most of the methods are focused on classification problems. Therefore, there is still room for further exploration of the application of MMML to prognosis challenges. Moreover, at the heart of MMML for PHM tasks is getting useful multi-modal latent representations using effective data fusion techniques. While simple concatenation methods are widely used for data fusion in MMML, there's a need for further exploration of more advanced fusion techniques, such as attention-based or transformer-based mechanisms. For future PHM data challenges, the provision of open-source multi-modal datasets would empower researchers to investigate and apply more advanced MMML techniques, potentially leading to significant advancements in the context of PHM.

5.3.2 A Lack of Adopting Systematic Approaches for Effective PHM Systems Construction

Our analysis of ML and DL methods across nine open-source industrial datasets has revealed the transformative potential that ML approaches can have on PHM. Nevertheless, the nature of the competition tends to prioritize solutions that chiefly enhance accuracy, potentially at the expense of systematic approach, reusability, and methodological inheritance. It is therefore vital to pursue systematic methodologies for constructing effective PHM systems that go beyond the competitive framework. This should involve thorough research and analysis of open-source datasets to advance ML and DL strategies, aiming not just for competition success but also for benchmarking and comparative analysis.

Souza et al. devised an ML-based, data-oriented pipeline for constructing a Prognosis and Health Management System (PHMS) focused on RUL prediction, utilizing semi-supervised ML with Autoencoder, XGBoost, and the SHAP method [99]. Professor Enrico Zio highlighted the primary challenges and future directions for implementing condition-based and predictive maintenance in real-world applications [127]. As shown in Figure 5, Hu et al. offered a new perspective on reviewing PHM efforts by proposing a division of the PHM lifecycle into DEsign, DEvelopment, and DEcision (DE^3) phases, and showcasing the pivotal activities and challenges within these stages [36]. Additionally, Lee et al. introduced a novel SoQ methodology for multi-stage manufacturing processes. It can help to analyze multi-parameter influences on product quality and model inter-process relationships in multi-stage manufacturing systems [58]. Additionally, Figure 6 shows an Industrial Large Knowledge Model framework proposed by Lee's team. This framework aimed to utilize the power of LLMs and domain-specific knowledge to address the complex challenges in the PHM domain [59].

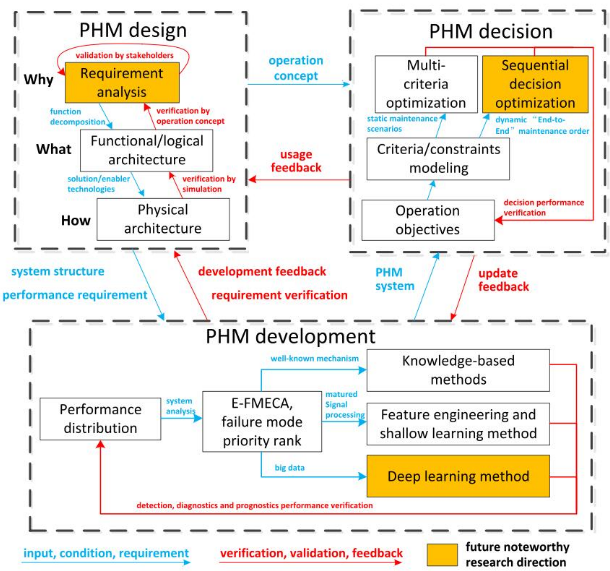

Figure 5. Interactions of PHM Phases of Design, Development, and Decision (DE^3) [36]

Moving forward, developing novel, systematic approaches for PHM systems should be encouraged in future PHM data challenge competitions. These efforts can notably augment the systematization and applicability of ML and DL approaches, thereby expanding their utility beyond the confines of the competition-centric paradigm. Such advancements promise to narrow the divide between academic research and industrial practice, facilitating the broader adoption of data-driven ML across various industrial contexts.

6 Prospects

There are still several pivotal suggestions that merit deeper investigation and exploration by the research community going forward.

(1) A Need for Open-Source Multi-Modal Datasets. In the PHM domain, the availability of multi-modal datasets is notably limited. While prior research has leveraged diverse modal information, including vibration, current, or torque signals for diagnosing issues in wind turbine gearboxes, bearings, or other industrial products [45, 114, 31], these datasets are all private. This restriction to some extent hampers the capacity for broad-based development of MMML approaches. To overcome this challenge, there is a need for the PHM community to collaboratively establish and maintain multi-modal industrial datasets, enriched with high-quality data and labels. This would involve the collection, alignment, and annotation of multi-modal data with PHM-centric attributes. Moreover, providing latent representations or pre-trained embeddings, if possible, can also accelerate and efficiently train new MMML models and facilitate knowledge transfer across various PHM tasks, ultimately benefiting the whole PHM community.

(2) Development of Multi-Modal Machine Learning Approaches. Furthermore, the investigation of a broader range of MMML techniques is highly encouraged. On the one hand, for time series data of a single modality, it may be feasible to extract features representative of different modalities from the time series data itself, such as the time domain (origin signal), frequency domain (FFT), PSD, STFT, etc. This approach could lead to the preliminary training of unimodal ML models on each single modality, followed by the exploration of MMML strategies. On the other hand, MMML methodologies necessitate more effective representation learning and information alignment techniques—the former concerning the efficient encoding of single modality data, and the latter focusing on the enhanced analysis and fusion of multimodal information for effective PHM prediction or classification tasks. Although simple concatenation is a common method for data fusion in MMML, emerging fusion techniques, such as attention-based or transformer-based mechanisms [111], deserve further exploration. These advanced methods have the potential to effectively capture implicit feature alignments across modalities and facilitate cross-modal synthesis [71, 119]. Yet, research on MMML within the PHM field remains underexplored, highlighting a significant opportunity for the adaptation of existing methods and the innovation of new approaches to address the unique challenges of PHM.

(3) Further Exploration in Machine Learning Model Interpretability. The adoption of DL in PHM has underscored the need for models that are not only high-performing but also interpretable. Techniques such as advanced data visualization and XAI methods are emerging as key tools in explaining the outputs of ML models [64, 6]. These methods are anticipated to provide industries and academia with clearer insights into the decision-making processes of PHM models, thereby building trust and facilitating more informed decision-making [98]. However, current XAI methods predominantly address the interpretability of models using tabular data, text, and images as inputs, leaving a gap in methods tailored for time series data. Moreover, most of the current interpretability methods are applied to unimodal ML models, and the interpretability of deep MMML models has not been explored. Addressing these gaps presents valuable opportunities for future research, aiming to balance the performance and interpretability of ML models. Such advancements would make ML models more trustworthy and interactive, enhancing their utility for scientists and engineers.

(4) Novel Transfer Learning and Domain Adaptation Techniques Development for Model Robustness, and Generalization. Alongside interpretability, the robustness and generalization of ML models in diverse and unpredictable industrial settings are paramount. Novel approaches in transfer learning and domain adaptation can be further developed to ensure models are resilient to data variability and operational uncertainties and capable of adapting to new, unseen scenarios [8]. This ensures their applicability in a wide range of PHM applications, driving the field towards more efficient, reliable, and transparent systems. Currently, research on transfer learning in the PHM domain predominantly addresses fault diagnosis, with only a few studies exploring prognosis. Looking forward, how to utilize the power of transfer learning for prediction problems is still a critical issue. Additionally, cross-modal transfer learning (CMTL) emerges as a critical area of interest in PHM, aiming to improve the knowledge transfer between distinct domains. Moreover, the challenge of collecting a sufficiently large, labeled dataset is a significant barrier in practical applications. The development of unsupervised and semi-supervised transfer learning techniques may help to address this issue.

(5) Construction of Large Knowledge Library. There have seen increasing applications of ML approaches in PHM, but the knowledge and solutions remain fragmented. Therefore, there is a need to establish a comprehensive large knowledge library that can include open-source industrial sensor data, problem-solving approaches, open-source code, model weights, technical papers, review documents, etc. This dynamic knowledge library could evolve over time, becoming more comprehensive and thereby benefiting the broader PHM community. By consolidating various industrial application information into this knowledge library, researchers can quickly access and utilize existing techniques when encountering familiar problems, thus accelerating the pace of research and enhancing problem-solving efficiency. For instance, Lee's team proposed to construct a industrial large knowledge library including human-interpretable data and structured machine generated data [59]. The proposed industrial LKL could serve as a domain-specific foundation for training Industrial Large Knowledge Models to solve domain-specific problems.

(6) Potential Utilization of Large Language Models (LLMs) and Industrial Large Knowledge Models (ILKMs). Recent advancements in large language model technologies have shown remarkable abilities in natural language processing and related tasks, hinting at the potential for general artificial intelligence applications [125]. In the PHM domain, leveraging these cutting-edge technologies could lead to groundbreaking changes. Yang et al. introduced a novel benchmark dataset focused on Question Answering (QA) in the industrial domain and proposed a new model interaction paradigm, aimed at enhancing the performance of LLMs in domain-specific QA tasks [116]. This approach signifies a substantial stride in customizing LLMs for more specialized, industry-oriented applications. Meanwhile, Li's team systematically reviewed the current progress and key components of ChatGPT-like large-scale foundation (LSF) models, and provided a comprehensive guide on adapting these models to meet the specific needs of PHM, underscoring the challenges and opportunities for future development [63]. Moreover, as shown in Figure 6, Lee's team proposed an Industrial Large Knowledge Model (ILKM) framework that aims to solve complex challenges in intelligent manufacturing by combining LLMs and domain-specific knowledge [59]. In summary, by integrating specialized domain knowledge with LLM technology, ML models could offer enhanced performance in addressing industrial data challenges specific to PHM.

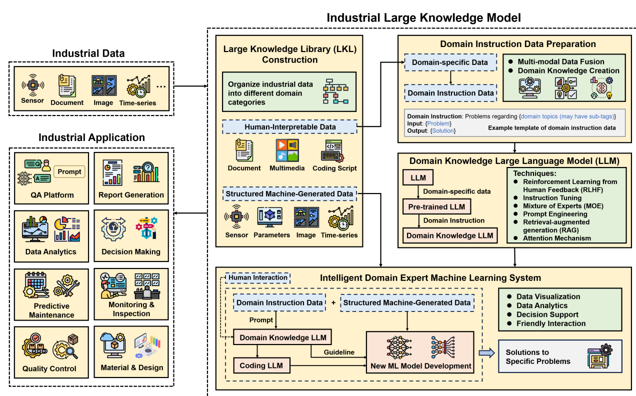

Figure 6. Industrial Large Knowledge Model Framework [59]

7 Conclusion

In summary, ML gradually becomes a cornerstone in PHM, reflecting the big potential for innovative advancements in future PHM development. This paper serves as a valuable resource for both academic and industry professionals in the PHM domain, offering a unified ML framework in PHM and a comprehensive overview of the current state-of-the-art ML approaches for diagnostics and prognostics of industrial systems using industrial open-source data. Based on two primary research task categories: ”Detection & Diagnosis” and ”Assessment & Prognosis”, we provide a detailed explanation of the problems, tasks, challenges, and relevant ML approaches to each competition. Furthermore, we summarize common challenges, including data-related issues and model-related issues, and analyze the solutions to address these challenges. Moreover, we evaluate the limitations of these PHM data challenge competitions and suggest future directions for the PHM data challenge competition evolution. Finally, we prospect six potential directions in the application of data-driven ML within PHM, encompassing a need for open-source multimodal datasets, development of MMML approaches, further exploration of ML model interpretability, improving the robustness, and generalization of ML models, constructing a large knowledge library, and harnessing the potential of LLMs and ILKMs.

Nomenclature

AI	Artificial Intelligence
$Bi-LSTM$	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
$DANN$	Domain Adversarial Neural Networks
DL	Deep Learning
DTW	Dynamic Time Warping
FCM	Fuzzy C-Means
FCN	Fully Convolutional Network
FIR	Feature Importance Ranking
GAN	Generative Adversarial Networks
GRU	Gated Recurrent Units
$ILKM$	Industrial Large Knowledge Model
KNN	K-Nearest Neighbors
LDA	Linear Discriminant Analysis
$LightGBM$	Light Gradient Boosting Machine
$LIME$	Local Interpretable Model-Agnostic Explanations
LKL	Large Knowledge Library
LLM	Large Language Model
$LSTM$	Long Short-Term Memory
ML	Machine Learning
MLP	Multi-Layer Perceptron
MMD	Maximum Mean Discrepancy
PCA	Principal Component Analysis
PDF	Probability Density Function
PHM	Prognostics and Health Management
PLS	Partial Least Squares
PSD	Power Spectral Density
QA	Question Answering
RNN	Recurrent Neural Network
RUL	Remaining Useful Life
$SHAP$	SHapley Additive exPlanations
$SMOTE$	Synthetic Minority Oversampling TEchnique
SoQ	Stream-of-Quality
$STFT$	Short-Time Fourier Transform
SVM	Support Vector Machine
SVR	Support Vector Regression
TCN	Temporal Convolutional Network
XAI	Explainable Artificial Intelligence
$XGBoost$	Extreme Gradient Boosting

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Appendix

Table 9. Task Description and Website Links for PHM Data Challenge Competitions

PHM Data Challenge Competition	Task Description	URL
2018 PHM NA Ion Mill Etching System	(1) Diagnose failures (2) Determine time remaining until next failure	https://phmsociety.org/conference/annual-conference-of-the-phm-society/annual-conference-of-the-prognostics-and-health-management-society-2018-b/phm-data-challenge-6/
2019 PHM NA Fatigue Crack	(1) Estimate crack length	https://data.phmsociety.org/2019datachallenge/
2020 PHM EU Filtration System	(1) Predict RUL of the filtration system	http://phmeurope.org/2020/data-challenge-2020
2021 PHM EU Manufacturing Production Line	(1) Fault detection (2) Fault diagnosis (Classification) (3) Root cause identification	https://github.com/PHME-Datachallenge/Data-Challenge-2021
2021 PHM NA Turbofan Engine	(1) RUL prediction in a fleet of aircraft engines	https://data.phmsociety.org/2021-phm-conference-data-challenge/
2022 PHM EU Printed Circuit Boards	(1) Prediction of defects in AOI based on SPI (2) Prediction of human inspection label (3) Prediction of human-assigned repair label	https://github.com/PHME-Datachallenge/Data-Challenge-2022
2022 PHM NA Rock Drill	(1) Multiclass fault classification for rock drills	https://data.phmsociety.org/2022-phm-conference-data-challenge/
2023 PHM AP Spacecraft Propulsion System	(1) Diagnose normal, bubble anomalies (2) Diagnose solenoid valve faults (3) Diagnose unknown abnormal cases	https://phmap.jp/program-data/
2023 IEEE Gearbox	(1) Fault diagnosis (Classification)	http://www.ieeereliability.com/PHM2023/index.html or http://www.ieeereliability.com/PHM2023/assets/data-challenge.pdf